Mambo

Track: Development — AI for Impact Challenge (AI4I 2026)

Project Title: Mambo — a whole-of-government citizen information assistant

Team Name: Team Mambo

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1. Problem Definition & Strategic Alignment

The problem. Zimbabwean citizens struggle to find reliable, plain-language government information. Official content is scattered across many ministry websites, frequently as scanned PDFs, written in legal jargon, and rarely answers the *whole* question — what to bring, what it costs, where to go. Citizens do not know which ministry handles what, so they make avoidable trips to the wrong offices. The cost is wasted time, uneven access, and frustration — sharpest for citizens on low-bandwidth mobile connections and for front-line registry staff who answer the same questions all day.

Who it serves. Citizens applying for IDs, passports, certificates and tax clearance; SME owners navigating business and tax registration; students and journalists verifying laws and policies; and ministry helpdesk/registry staff who need consistent, cited, first-line answers. A mobile-first, free, always-on assistant reaches the widest audience.

What Mambo is. A retrieval-augmented assistant that answers in plain language using **only** an allow-listed corpus of official documents, with a citation on every claim, an honest **evidence-status badge** on every answer, structured **service-journey cards** for common tasks, and a **ministry handoff card** when it cannot answer. *Mambo* is a Shona word for "matters / affairs" — fitting for a service that handles the public's business.

Strategic alignment. Mambo implements the National AI Strategy's goals for inclusive, trustworthy public-service AI built on local capability and digital sovereignty: official-source grounding with citations (trustworthy); abstention and honest status (responsible); mobile-first free access (inclusive); a self-hostable open-source stack with a CCE roadmap (sovereign, local). It is whole-of-government — one brain, many ministries — led by ICT and designed so every ministry plugs in via a curated registry.

2. Technical Design & Product Logic

Architecture — three modules on different rhythms, meeting at one Knowledge Store. Splitting by rhythm lets each scale independently:

```
 ingestion/ (batch)  ── discover · OCR · chunk · embed ──▶  PostgreSQL + pgvector
                                                              (Knowledge Store)
 webchat/  ──/api──▶  rag/ (FastAPI, per question)
──route·retrieve·generate·trust──▶
 (Next.js 14)                                                      reads ▲
```

Per-question pipeline: `guard → intent → route → retrieve → (confident?) → generate or fall back → trust layer`. The safety guard abstains before retrieval (no LLM needed); generation sits behind a **swappable** OpenAI-compatible interface.

Models and stack.

Component	Choice	Why
Generation	DeepSeek (flash), swappable	Phase 2 self-host an open-weight model on the CCE — no code change
Embeddings	Qwen3-Embedding-8B (4096-dim), self-hosted (Ollama)	Multilingual → Shona/Ndebele is an addition, not a re-embed

Component	Choice	Why
Knowledge store	PostgreSQL 16 + pgvector	Runs on a cheap cloud box *and* a Ministry server — one tech, two homes
Retrieval	Ministry-scoped cosine + recency boost	Newest version of an amended law wins
Webchat	Next.js 14, streaming SSE, mobile-first	Low-bandwidth, always-on

Data. 903 official documents / 3,059 chunks from 9 allow-listed sources (5 ministries + ZIMRA, ZIMSEC, Veritas, ZimLII). Every chunk stores source URL, page, fetch date, content hash and raw path — so every citation is verifiable.

Why AI is necessary (not a sledgehammer). Plain-language synthesis, multi- ministry routing, conversational follow-ups ("what about for schools?"), and cited answers over many long documents cannot be met by keyword search, static FAQs, or SQL. RAG is the appropriate, minimal use of AI: retrieve official passages and ground generation in them. We do not apply AI where a rule would do.

Product features (built). Ministry routing ("find the right office"); inline `[n]` citations on every answer; honest **evidence-status badge** (answered / partial / unsupported / declined); a deterministic **abstention guard** (medical, legal, personal-data, political, prompt-injection); **service-journey cards** for six common tasks; **ministry handoff cards**; streaming; mobile-first; light/dark themes.

Measured results (not asserted). A fresh build from a clean checkout boots; **59 automated tests pass**. On the corpus and a 32-question citizen-query set:

Metric	Result
Router accuracy (English)	89.3% routed to an expected ministry
Dangerous-case abstention	7 / 7 handled correctly
Corpus integrity checks	8 / 8 pass
Citation link liveness (sample)	8 / 8 resolved
"Official-sources-only" claim	True (web verifier off by default)

Honest gaps are disclosed, not hidden: education (3 chunks) and ZimLII (1) are thin; Shona/Ndebele router accuracy is 60% (Phase 2; embeddings are already multilingual).

3. Deliverables & CCE Implementation Roadmap

What is delivered. A working end-to-end MVP (ingestion → RAG → webchat), deployed and demonstrable; a reproducible test and evaluation harness; and a full submission evidence pack (data governance, deployment, business, design, evaluation).

Timeline (8 weeks of AI4I support).

Weeks	Workstream	Output
1–2	Harden	Close remaining security/schema/test items; run full RAG eval on resourced

Weeks	Workstream	Output
		compute; fix accessibility “pending” items
3–4	Expand	Close coverage gaps (re-crawl education; expand ZimLII); add Shona/Ndebele registry keywords; measured sn/nd pilot
5–8	Pilot	Home Affairs pilot: confirm sources + handoff contacts, reviewed-answer cache, controlled user group, weekly review, go/no-go

ZCHPC CCE roadmap. The stack is CCE-ready by design (open-source, self-hostable, swappable model interface). On the CCE we run PostgreSQL+pgvector (the Knowledge Store), bulk embedding jobs on GPU (`ingestion/embed_bulk.py` already targets a remote GPU Ollama endpoint), a self-hosted open-weight generation model behind the swappable interface, and the evaluation pipeline. Migration is a re-point of `OLLAMA_BASE_URL` / `DEEPSEEK_BASE_URL` — no code change.

CCE milestone	Outcome
Provision	CCE partition: Postgres+pgvector + Ollama/vLLM endpoints
Migrate	Move the Knowledge Store to CCE (dump/restore; same schema)
Re-embed	Bulk embeddings on GPU; run full evaluation
Benchmark	Latency/cost vs current local+API setup; publish metrics
Operate	Incremental refresh + monitoring from CCE

This removes per-query API cost and keeps data in-country. **Mambo is cloud-deployed today, not edge**; the CCE is the documented scale/sovereignty path, and mobile-first low-bandwidth is the user-facing strategy.

4. Compliance & Risk Mitigation

Data Protection Act [Chapter 12:07].

Obligation	How Mambo handles it
Lawful basis	Public official documents (low personal-data risk)
Data minimisation	Query log keeps only needed fields; `client_ip`/`user_agent` retention-bounded
Retention & deletion	Defined retention window + deletion workflow (planned PII redaction)
Access control	Restricted log access by role
Transparency	User notice: do not enter ID/medical/private information
Open-data release	**Not** claimed for query logs

Cybersecurity. Nonce challenge (single-use) on write endpoints; origin lock to the official app; per-IP rate limiting + concurrent-stream cap; bot user-agent blocklist + behaviour/entropy checks; validated input sizes; docs/redoc disabled in production; secrets never committed. The deterministic abstention guard and official-sources-only retrieval harden the content layer. (59 tests cover nonce, abstention, and security behaviour.)

Responsible AI. Mandatory citations; a confidence threshold below which Mambo does not guess; the evidence-status badge; abstention for out-of-scope/unsafe questions with a referral (clinician / Law Society / ministry); per-section "not specified" in journeys; recency boost for amended laws; a risk register and monitoring plan with drift detection (PSI/KL) and a rule-baseline fallback.

Key risks (full register in `submission/deployment/risk_register.csv`): hallucination (grounding + evidence status + abstention); coverage gaps (disclosed, being closed); latency on local hosting (CCE roadmap); local-language inequity (honestly disclosed, Phase 2); PII in queries (minimisation + redaction plan).

5. Sustainability & Future Adoption

Cost reality. Infrastructure is inexpensive (~USD 50/month at pilot); the real cost is people — a reviewer per ministry batch and part-time engineering (`submission/business/cost_model.csv`). The stack is open-source and self-hostable, so a Ministry can run it on its own servers with **no per-query API cost** once on the CCE.

Adoption path. Pilot with one ministry (Home Affairs proposed) under AI4I milestone support → per-ministry onboarding (repeatable, via `ministry_onboarding_pack.md`) → whole-of-government rollout (the registry is the keystone) → embeddable widget on live ministry sites + low-bandwidth channels (WhatsApp).

Maintainability. Incremental refresh re-ingests only changed documents; a quarterly re-verification cadence; a reviewed-answer cache for the top ~50 questions per ministry that doubles as a rule-baseline fallback; monitoring that drives what to fix next.

A note on fallibility. Mambo is a **guidance and routing assistant**, not a replacement for official decisions, professional advice, or ministry confirmation. It points citizens to the right office and the right documents, and it says so honestly when the official record does not cover a question. That scope discipline is how it earns trust.